Drug Response Prediction in Leukemia Cell Lines

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I. Individual Contributions

Marchelle Le: Data preparation, code contribution (naive bayes classifier: naive_bayes_classifier.py, run_naive_bayes_classifier.py, run_naive_doxo.py, run_naive_navi.py, run_naive_combo.py; regression tree: regression_tree.py, run_regression_single.py, run_full_lasso_decisiontree_summary.py, un_regression_tree_with_lasso.py), code review, writing contribution (sections: introduction, methods, results)

Mantasha Noyela: Code contribution (regression tree: run_full_lasso_decisiontree_summary.py, run_regression_tree_with_lasso.py, run_lasso_combined.py, run_lasso_doxorubicin.py, run_lasso_navitoclax.py), writing contribution (sections: discussion, conclusion)

Natasha Black: Data preparation, code contribution (naive bayes classifier: naive_bayes_classifier.py, run_naive_combo.py; regression tree: regression_tree.py, run_regression_tree_with_lasso.py, run_full_lasso_decisiontree_summary.py, run_regression_combo.py, run_lasso_combined.py, run_lasso_doxorubicin.py, run_lasso_navitoclax.py), code review, writing contribution (sections: introduction, data, methods, results, discussion)

II. Introduction

The purpose of this project focuses on predicting how leukemia cancer cell lines respond to specific anticancer drug treatments using machine learning techniques. Leukemia is a type of cancer characterized by the overproduction of abnormal white blood cells in the bone marrow (Chennamadhavuni et al. 2023). Leukemia treatments include chemotherapy drugs that vary in effectiveness based on the patient's genetic and molecular profile. A key focus in oncology is understanding how gene expression affects drug response. Using methods such as RNA sequencing allows researchers to measure gene expression across thousands of genes in cancer cells and identify which genes correlate to resistance and sensitivity to specific drugs. In addition with drug response measurement known as Area Under Curve values (AUC) as it shows the rate of sensitivity of a cancer cell line to a drug we use gene expression features derived from RNA sequencing dataset, in this experiment.

Computational oncology has a core problem of predicting response for anticancer drugs on cancer cell lines using molecular profiles (Bunne et al. 2023). It is a critical step towards the development of personalized medicine. Machine learning techniques in this project involve the use of a regression tree and a Naive Bayes classifier. These models are evaluated based on

prediction performance of a cell line's response to single agent and combination drug treatments. Binarized drug treatment indicators are incorporated to enable combination testing. For the regression tree, metrics such as RMSE and R^2 are used to evaluate the model, whereas for the classifier, metrics such as accuracy, precision, recall, and F1 score are used to evaluate the model.

III. Data

The dataset used in this project is publicly available and stored at the Predictive Oncology Model and Data Clearinghouse as the Pilot 1 Cancer Drug Response Prediction Dataset, which combines cell line drug response data from a collection of other studies (Basu et al. 2013; Heath et al. 2021; Holbeck et al. 2017; Gandhi et al. 2019; Rees et al. 2016; Seashore-Ludlow et al. 2015; Smirnov et al. 2018; Subramanian et al. 2017; Yang et al. 2013). Sources that it has compiled data from include the Cancer Cell Line Encyclopedia (CCLE), Genomics of Drug Sensitivity in Cancer (GDSC), and NCI-60. Included in this compilation of data is gene expression data from bulk RNA sequencing. Drug response data was reported through various means, including AUC, wherein a low value indicates higher sensitivity to a given drug or combination of drugs. Given how large the transcriptome is, a subset of the gene expression data will be used for prediction, including only genes measured across all cell lines and from the LINCS1000 gene set.

Only a few drugs, with one combination, will be chosen for training and prediction. The single agents chosen include Doxorubicin and Navitoclax, while the combination for prediction will be Doxorubicin:Navitoclax. Given that there are many distinct cancers that have been screened in this study, we will be evaluating the subset of leukemia cell lines in the data. After filtering for just the leukemia cell lines, only 505 samples remained. Not every sample has drug response data from each of the drug treatments, with 72 samples having AUC values for Navitoclax, 203 samples having AUC values for Doxorubicin, and 203 samples having AUC values for the combination.

IV. Methods

A. Regression Tree

The regression tree algorithm recursively partitions the gene expression features to reduce the variance in the AUC value. Each node represents a gene and a threshold for expression. Each leaf node has the average AUC value of the related subset of samples. Before training the tree, Lasso regression is applied to select a subset of predictive genes from RNA sequencing data, which is crucial for preventing overfitting and improving interpretability of the tree outcome. Separate Lasso models were fitted for Doxorubicin, Navitoclax, and the Doxorubicin:Navitoclax combination, each producing a unique feature set.

The gene expression is standardized using z-score normalization once the Lasso features are selected. Next, the data was split 80/20 into training test sets. The limitations of this recursive algorithm are the number of samples at a node is less than 5, the maximum depth of tree of 5 is reached, and the standard deviation of AUC in a node is less than 0.15. In addition, pruning is applied to further reduce overfitting, which is controlled by regularization parameter α.
Next, the RSME and R^2 are calculated to analyze the performance. The algorithm was applied independently to Doxorubicin, Navitoclax, and the Doxorubicin:Navitoclax combination. Lastly, the regression tree with Lasso prints out its tree structure and plotted from matplolib function, to visualize its decision paths.

B. Naive Bayes Classifier

For the naive bayes classifier, the feature values were discretized to three levels, with 0 corresponding to low gene expression, 1 corresponding to intermediate gene expression, and 2 corresponding to high expression of a gene. The target AUC values were also discretized, such that a low AUC between 0 and 0.33 (corresponding to high response to a drug) corresponds to 2, moderate AUC 0.34-0.66 corresponds to 1, and a high AUC of 0.67-1 (low drug response) corresponds to 0. Additionally, since not all drugs have response data for a given treatment, binary columns for whether a cell line has been treated with a drug have been added to the feature matrix. These binary columns are for use in the training of the models for combination response prediction, but they won't be used for single agent response prediction.

The naive bayes classifier first calculates the prior probabilities of the training data. Then it goes through the training data, creating a dictionary that contains instances of classes corresponding to particular feature values, similar to the weather example from the bayes net slides for the course. This dictionary is used with the test data for laplace smoothing, such that new instances can be calculated. After calculating the probability of the new observation being assigned to a given class, the class with the maximum probability is chosen and assigned to the new observation. The results of the predicted and actual drug response classes were displayed in a confusion matrix and evaluated with the metrics of accuracy, precision, recall, and F1 score.

V. Results

A. Regression Tree

1. Doxorubicin

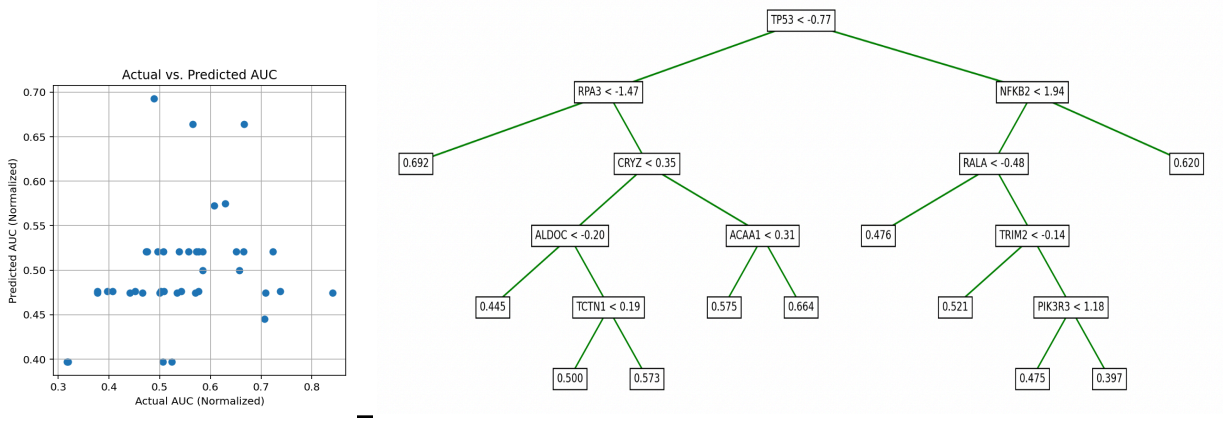


Figure 1. Scatter plot (left) of predicted and actual AUC for Doxorubicin and regression tree (right) with Lasso-selected features for Doxorubicin.

Evaluation Metric	Value
RMSE	0.1220
Test R^2	-0.1498

Table 1. Table of evaluation metrics for Doxorubicin regression tree.

Features selected by lasso regularization:
ABHD6, ACAA1, ADAT1, ADRB2, ALDOC, ANO10, ATP2C1, ATP5S, ATP6V1D, BAMBI, BAX, BECN1, BZW2, CASP3, CCL2, CD320, CD44, CIRBP, CORO1A, CPNE3, CRK, CRYZ, CSNK1E, CSRP1, DAG1, DCUN1D4, DDR1, DHRS7, DNAJC15, DUSP3, EBNA1BP2, ECH1, EDN1, EPHA3, EVL, EXT1, EZH2, FAM20B, FHL2, FOSL1, FSD1, GADD45B, GALE, GHR, GLI2, GPC1, HOMER2, HSPD1, IKZF1, KCTD5, KIT, MALT1, MAPK1IP1L, MAST2, MAT2A, MMP2, MSRA, MTA1, MYBL2, NFIL3, NFKB2, NIPSNAP1, NUP62, PAK4, PAN2, PGAM1, PIK3C2B, PIK3R3, PLA2G15, POLB, POLE2, PRCP, RALA, RBM34, RFNG, RPA3, RPP38, RPS6, RRAGA, S100A4, SCCPDH, SERPINE1, SH3BP5, SLC11A2, SLC25A14, SLC27A3, SLC5A6, SMC3, SPP1, STUB1, STX1A, TBP, TCTN1, TES, TEX10, TGFBR2, TJP1, TLK2, TNFRSF21, TOMM34, TP53, TRAK2, TRAPPC6A, TRIM2, TSPAN3, TSPAN6, UBE2C, UBE3B, ZNF395, ZNF586

2. Navitoclax

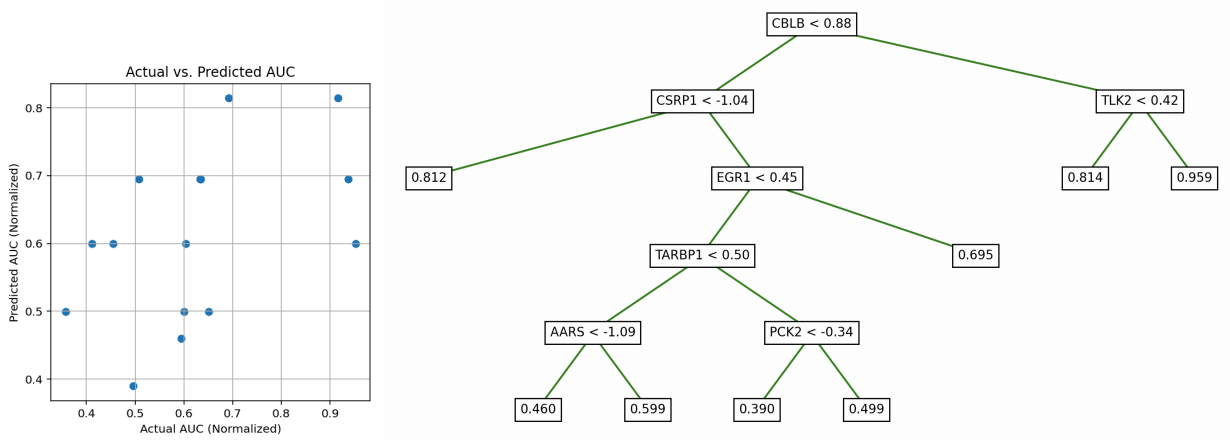


Figure 2. Scatter plot (left) of predicted and actual AUC for Navitoclax and regression tree (right) with Lasso-selected features for Navitoclax.

Evaluation Metric	Value
RMSE	0.1615

Test R^2	0.1748

Table 2. Table of evaluation metrics for Navitoclax regression tree.

Features selected by lasso regularization:

AARS, ACAA1, ACBD3, ADH5, ADRB2, BACE2, BAD, BAMBI, CASP10, CBLB, CERK, COG7, CRYZ, CSRP1, CTNNAL1, DNAJA3, DNAJC15, EGF, EGR1, EPHA3, EVL, FBXO21, FGFR4, FYN, GADD45A, GAPDH, GHR, GNA15, GNAI2, HDAC2, HOMER2, HOXA5, KCNK1, KCTD5, KDM3A, KLHL9, LIPA, MAMLD1, MAP3K4, MAPKAPK3, MMP1, NOS3, NRIP1, PCCB, PCK2, PEX11A, PNP, POLE2, PPP2R3C, PRCP, PTGS2, PTK2B, PTPRK, PXN, RGS2, RPP38, SOCS2, SPAG4, SYPL1, TARBP1, TCTN1, TFDP1, TLK2, TRIB3, TRIM2, TXNDC9, UBE3B, UFM1, ZMIZ1

3. Combination Treatment

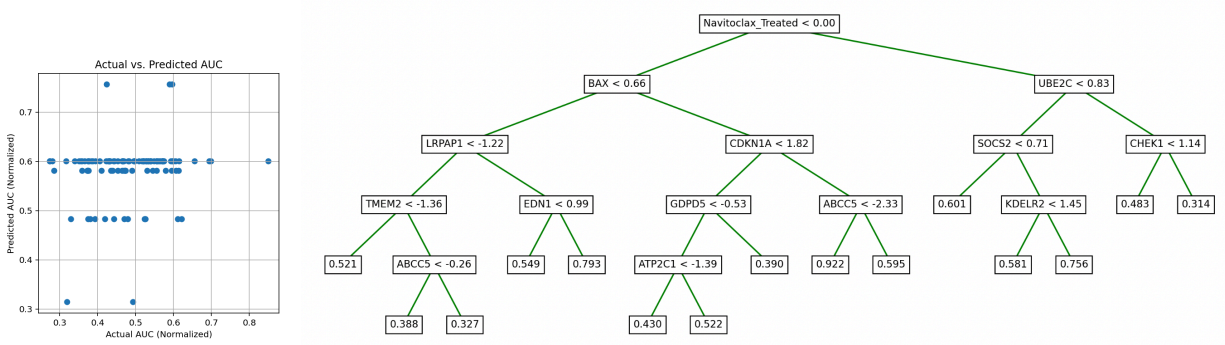


Figure 3. Scatter plot (left) of predicted and actual AUC for Doxorubicin:Navitoclax combination treatment and regression tree (right) with Lasso-selected features for Doxorubicin:Navitoclax.

Evaluation Metric	Value
RMSE	0.1471
Test R^2	-1.0981

Table 2. Table of evaluation metrics for Doxorubicin:Navitoclax combination treatment regression tree.

Features selected by lasso regularization:

ABCC5, ADO, ALDH7A1, ARFIP2, ATF6, ATP2C1, ATP6V0B, ATP6V1D, BAX, CASP3, CBR1, CDK1, CDKN1A, CHEK1, CHP1, CIAPIN1, CIRBP, COL4A1, CORO1A, CPNE3, CSRP1, CXCL2, DCUN1D4, DHRS7, DNM1L, DNMT3A, DUSP3, EDN1, EGFR, FAM20B, FAT1, FSD1, GDPD5, GPC1, HIST1H2BK, ICAM3, IKZF1, INSIG1, ITGB5, KDELR2, KEAP1, KIF5C, KIT, LIPA, LRPAP1, LSR, MALT1, MAP4K4, MCOLN1, ME2, MMP1, MTA1, NENF, NIPSNAP1, NR1H2, NUP62, OXCT1, PAK1, PAX8, PCM1, PDGFA, PHKA1, PIK3C2B, PIK3R3, PLSCR1, POLE2, PSMD4, PSMD9, PTGS2, PTK2, PTPN1, RAD51C, RFNG, RUVBL1, S100A4, SCAND1, SENP6, SERPINE1, SH3BP5, SIRT3, SLC1A4, SLC35A1, SLC35B1, SLC5A6, SNX7, SOCS2, SPP1, STAT3, STXBP2, TFAP2A, TJP1, TMEM2, TNFRSF21, TOMM34, TP53, TRAK2, TSPAN3, TSPAN6, UBE2C, UBE3C, VGLL4, WRB, XPNPEP1, ZNF586, ZNF589, Navitoclax , Doxorubicin

B. Naive Bayes Classifier

1. Doxorubicin

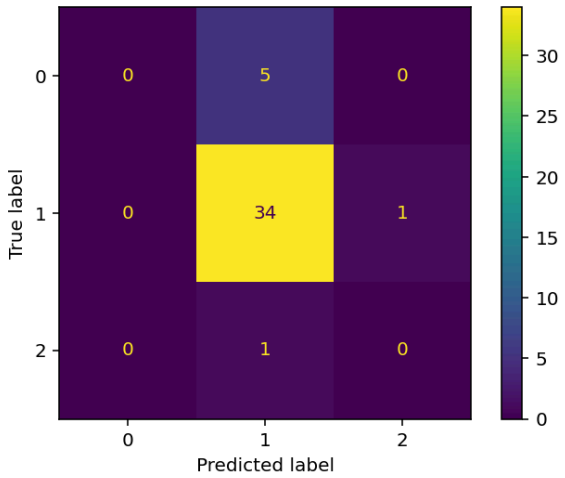


Figure 4. Confusion matrix of predicted and actual drug response class for Doxorubicin.

Class	Precision	Recall	F1	Total cases
0	NaN	0	0	5
1	0.85	0.97	0.91	35
2	0	0	0	1

Table 4. Table of evaluation metrics drug for classification of drug response with Doxorubicin.

Overall Accuracy: 83%

2. Navitoclax

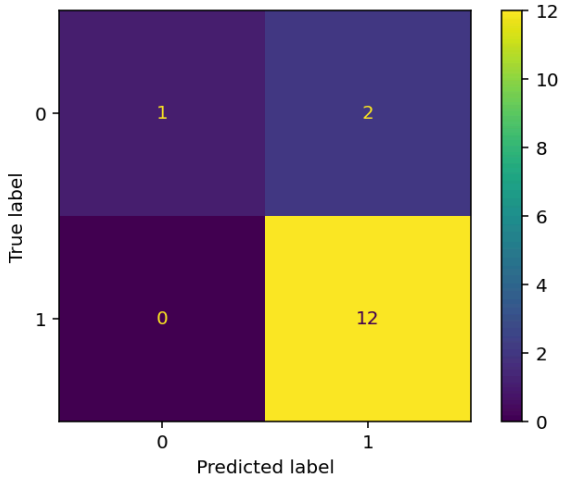


Figure 5. Confusion matrix of predicted and actual drug response class for Navitoclax.

Class	Precision	Recall	F1	Total cases
0	1	0.33	0.5	3
1	0.86	1	0.92	12

Table 5. Table of evaluation metrics drug for classification of drug response with Navitoclax.

Overall Accuracy: 87%

3. Combination Treatment

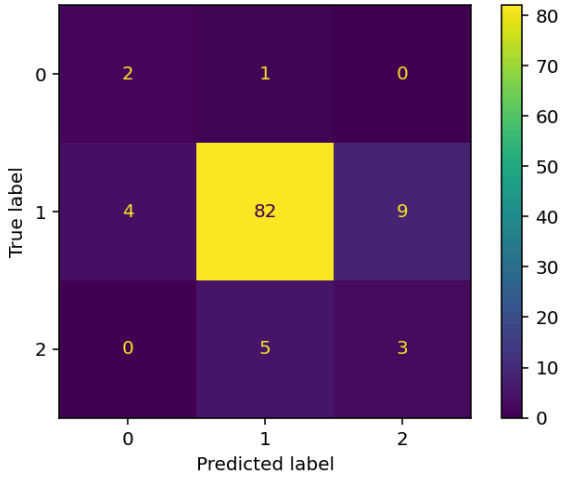


Figure 6. Confusion matrix of predicted and actual drug response class for Doxorubicin:Navitoclax combination treatment.

Class	Precision	Recall	F1	Total cases
0	0.33	0.67	0.44	3
1	0.93	0.86	0.90	95
2	0.25	0.38	0.3	8

Table 6. Table of evaluation metrics drug for classification of drug response with Doxorubicin:Navitoclax combination treatment.

Overall Accuracy: 82%

VI. Discussion

A. Decision Tree

The regression tree models, regularized with Lasso-based feature selection, displayed varying levels of predictive performance across three treatment scenarios. For Doxorubicin, the model achieved a relatively low root mean square error (RMSE) of 0.1220. However, the negative R^2

value of -0.1498 indicates that the model performed worse than a baseline mean predictor, reflecting poor generalization. In contrast, the Navitoclax model showed a higher RMSE of 0.1615 but a positive R^2 of 0.1748, suggesting that it was somewhat more effective in capturing the underlying response patterns. The performance of the model significantly declined for the combination treatment of Doxorubicin and Navitoclax, which resulted in a modest RMSE of 0.1471 but an extremely negative R^2 of -1.0981, indicating a substantial deviation from the actual AUC values. Scatter plots supported these findings, revealing tighter clustering for the single agents and greater dispersion for the combination therapy.

The decision trees, even after Lasso regularization, were able to identify key genes in leukemia as decision points for regression. For example, in the decision tree for Doxorubicin, the tree identified NFKB2 expression as a split point, and there are studies that point to its pathway becoming dysregulated in some cancers, leading to Doxorubicin resistance (Kim et al. 2016). Given this context, it's interesting that the decision tree was able to identify this as a split point for drug response. Additionally, treatment with navitoclax was an important split point in the decision tree for the combination treatment prediction. Of note, one of the main nodes branching off is BAX, which is a part of the BCL2 family of genes, with this gene family being a target of Navitoclax (Lopez et al. 2022). Despite the model's performance, it was interesting that it was able to recover genes relevant to leukemia and to the drugs chosen for evaluation in this project. Despite these drawbacks, the regression trees identified biologically meaningful genes, such as NFKB2 and BAX, which are associated with known mechanisms of drug resistance and apoptosis. This underscores the interpretative advantages of regression trees, even when their predictive performance is limited.

B. Naive Bayes Classifier

Looking at the results of the naive bayes classifier, it would seem to perform surprisingly well, just based on the accuracy metric alone, and given the complexity of drug response prediction. However, the accuracy is misleading, as there is a very large class imbalance in the data that leads the classifier to become very good at predicting a moderate drug response, which leads to it achieving an accuracy of 83% for Doxorubicin, 87% for Navitoclax, and 82% for the combination treatment, just by predicting that the majority of the cases would have a moderate response to the drug, since that is the main class present.

However, having a more in-depth look at the results, for predicting the response of these cell lines to Doxorubicin treatment, the classifier failed to classify any of the cases correctly as having resistance to the drug (class 0) or being sensitive to the drug (class 2). While there are cases of resistance and sensitivity in both the training and test data for this drug, the majority of cases in both sets are the moderate response class (class 1). Looking at the prediction results of Navitoclax the accuracy improves, but the full set of training and test data is even smaller here,

with only 72 cases and no examples of drug sensitivity, leaving only 2 classes. This does allow the model to successfully classify 1 case of resistance, but the remaining 2 resistant cases are classified as moderate response. This leads to low (but not 0 this time) recall and F1 scores for the resistant class.

Finally, taking a look at the prediction results for the combination drug treatment, the accuracy is deceiving. Again, it predicts most of the moderate response cases correctly, but most cases outside of that class are misclassified as moderate response. This time, even a fair number of moderate response cases were not classified correctly. This seems to suggest that the model was actually learning something from the data and not just overfitting. Unlike the Doxorubicin case, some of the resistant and sensitive cases were actually correctly classified. The precision, recall, and F1 metrics for these classes are still low, but it seems unexpected that they would be higher than for the single agent instance. However, these higher scores seem to be due to having more cases of the sensitive and resistant classes in the training and test data for this prediction, allowing for more chances to learn those classes and more opportunities to classify them.

Having a more balanced set of classes to predict could have helped to improve the predictive capabilities of the model in these cases, but unfortunately, small datasets with few examples of successful response (and many examples of relapse) are a common issue in this domain. This becomes even more different when moving from comparatively simple biological models, like cell lines, to real patient samples.

C. Comparison

When comparing the two methods, it seems as though the decision tree methods performed a bit better, with the single agent plots following somewhat of a linear trend. It may not have performed the best, but it seems better than the single agent Bayes classifier only being good at classifying cases of moderate response. For the combination treatments though, it seems like the naive Bayes classifier improved, not in accuracy but in respect to the other performance metrics, due to having more response cases to learn from, whereas the performance of the decision tree was quite poor for the combination treatment.

VII. Conclusion:

This project demonstrated the use of machine learning techniques, specifically regression trees combined with Lasso regularization, to predict drug responses in leukemia cell lines using gene expression data. While regression trees provided interpretable models and identified biologically relevant features, their predictive performance varied significantly across different drugs. The model showed moderate success in predicting responses to single agents, particularly for

Navitoclax, suggesting that it has some capacity to capture meaningful patterns. The Naive Bayes classifier demonstrated unexpectedly high accuracy scores across all treatments. However, a more in-depth analysis exposed its limitations, mainly due to significant class imbalance. It tended to predict moderate response classes predominantly, while struggling to accurately identify resistant or sensitive cases. The difficulty in predicting combination treatment responses across both methods highlights the challenges associated with modeling more complex biological interactions, especially given the limited data available.

Despite these limitations, the integration of Lasso feature selection with decision trees creates a valuable framework for biomarker identification, offering insights that are both computationally manageable and biologically interpretable while Naive Bayes offers simplicity and efficiency. Future work should focus on increasing the sample size, employing more robust ensemble methods, and leveraging additional biological knowledge to enhance both prediction accuracy and clinical relevance.

VIII. Citations

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